

Extraction Lab Biomass Flow

Step-by-step process from charging a biomass lot into extraction through the first wet THCA and HTE pour / output confirmation.

BUILT FROM

Implemented extraction app workflow, run progression state machine, run form fields, and booth SOP alignment notes in this repo.

AUDIENCE

Extraction operators, shift leads, supervisors, and production trainers.

FORMAT

Landscape workflow table with process name, responsible role, and time / timing expectation.

BOUNDARY

Ends when wet THCA and wet HTE are poured / weighed and initial outputs are confirmed. Later drying, milling, terp stripping, and destination queues are outside this document.

This is an operational training map inferred from the current software and planning documents. It is not a replacement for the approved safety SOP, equipment manuals, or state compliance requirements.

How To Read This

The rows are ordered in the sequence the team should experience the run. Branches are labeled where the workflow can split or loop.

-  **App checkpoint:** the current app has a button, field, timer, or stored timestamp for this.
-  **Manual / inferred:** the software implies this operational work but may not capture it as its own step.
-  **Critical control:** prerequisite, measurement, evidence, or safety-sensitive decision.
-  **Branch / loop:** the process changes direction based on an operator decision.
-  **Handoff:** the extraction run closes and wet outputs are transferred to the downstream process.

Suggested roles use the app's apparent launch-role model: extractor/operator, supervisor/shift lead, downstream operator, and maintenance/facilities where relevant.

Phase 1 - Charge, Reactor Prep, And Primary Extraction

Biomass lot enters extraction and primary solvent contact is completed

STEP	NAME OF PROCESS	PERSON / ROLE RESPONSIBLE	TIME	WHAT HAPPENS / CRITICAL NOTES
Lot selection and charge creation				
1	Scan or enter biomass lot APP	Extractor / operator	Start of extraction session	Operator scans or types the tracking ID in the standalone extraction app. The app opens the lot charge workflow and shows source, supplier, strain, and available biomass context.
2	Confirm lot is ready to charge CRITICAL	Extractor; supervisor if exception	Before biomass is loaded	Confirm the lot is available, correct, and ready. The app has ready/not-ready state and charge warnings. Do not proceed if the selected lot, weight, or reactor is wrong.
3	Record extraction charge APP	Extractor / operator	When biomass is committed to a reactor	Choose reactor and charge weight. The current app favors 100 lb, Half lot, Full lot, or Last used presets. Recording the charge links the lot, reactor, and draft run.
4	Physically load biomass / baskets into reactor INFERRED	Extractor / operator	Immediately after charge is recorded	Load the charged biomass into the selected reactor using site SOP. App stores charge weight, fill count, total fill weight, stringer basket count, and blend %, but does not currently spell out every physical loading movement.
5	Confirm reactor/tank readiness INFERRED	Extractor; maintenance if needed	Before vacuum down	Confirm reactor, recovery path, filtration path, and relevant tanks are clean, assembled, closed/open as required, and safe to pull down. This is critical but not fully modeled as a separate app checkpoint.
Primary extraction				
6	Confirm vacuum down CRITICAL	Extractor / operator	Before solvent charge	App button: Confirm Vacuum Down. This records timestamp/operator and is required before solvent charging can continue.
7	Prepare solvent supply / tank charge path INFERRED	Extractor; maintenance/facilities as assigned	Before or during solvent charge setup	Confirm solvent source/tank is ready and valve path is correct. The app captures solvent charge amount, not the full equipment prep sequence.
8	Record primary solvent charge CRITICAL	Extractor / operator	After vacuum; before soak	App field/button: Primary Solvent Charge, typically 500 lb per placeholder/default language. Weight must be positive. Timestamp is stored when Record Solvent Charge is pressed.
9	Start primary soak APP	Extractor / operator	Target: 30 min	App button: Start Primary Soak. Timing target is settings-driven and currently defaults to 30 minutes. Timing misses can warn, require override, or hard stop depending on settings.
10	Start mixer / agitation APP	Extractor / operator	During primary soak; target: 5 min	App button: Start Mixer. The mixer stage is timed separately so the operator can document agitation during the soak.
11	Stop mixer APP	Extractor / operator	After agitation target	App button: Stop Mixer. This completes mixer timing and advances the operator to the filter-clear checkpoint.
12	Confirm basket filter clear CRITICAL	Extractor / operator	After mixer; before pressurization	App button: Confirm Filter Clear. The SOP plan describes this as clearing the basket filter before nitrogen pressurization.

STEP	NAME OF PROCESS	PERSON / ROLE RESPONSIBLE	TIME	WHAT HAPPENS / CRITICAL NOTES
13	Start nitrogen pressurization CRITICAL	Extractor / operator	After filter clear	App button: Start Pressurization. This begins the pressure step that moves the process toward recovery.
14	Begin recovery / flow to filtration and recovery APP	Extractor / operator	After pressurization begins	App button: Begin Recovery. This checkpoint represents sending gas/liquid through filtration and recovery. Heat-exchanger handling is implied by the SOP plan but not currently its own app button.

Phase 2 - Flush Cycle, Recovery, And Final Purge

Critical temperature, solvent, flow, clarity, and shutdown controls

STEP	NAME OF PROCESS	PERSON / ROLE RESPONSIBLE	TIME	WHAT HAPPENS / CRITICAL NOTES
Transition from primary extraction into flush				
15	Begin flush cycle APP	Extractor / operator	After primary recovery checkpoint	App button: Begin Flush Cycle. This moves the run from primary extraction checkpoints into flush controls.
16	Verify flush solvent and plate temperatures CRITICAL	Extractor / operator; supervisor if out of range	Before flush solvent charge	App fields: Flush Solvent Chiller Temp and Plate Temp. Placeholder indicates -40 F or below for solvent chiller. Current app also supports evidence uploads for solvent chiller and plate temp photos.
17	Post / document temperature evidence PARTIALLY CAPTURED	Extractor / operator	With temperature verification	The app has a Posted to Slack checkbox and photo evidence counts. Slack reliability was previously deferred, so the app should be treated as the stronger system record for temp proof.
18	Record flush solvent charge CRITICAL	Extractor / operator	After temp verification	App field/button: Flush Solvent Charge, typically 500 lb per placeholder/default language. Requires temp verification in the guided flow.
19	Start flush soak APP	Extractor / operator	Target: 10 min	App button: Start Flush. Timing target is settings-driven and defaults to 10 minutes. The run also stores flush count and total flush weight.
20	Flush agitation / valve handling INFERRED	Extractor / operator	During flush soak	The SOP plan calls for flush soak with mixer/agitation, then burp and re-pressurization. The app currently captures the flush timer and surrounding checkpoints more clearly than each physical valve movement.
21	Stop flush APP	Extractor / operator	After flush target / when flush is complete	App button: Stop Flush. This closes the flush timer and advances the process to the flow-resumed decision.
Branch A/B - flow resumed decision				
22A	Branch A - flow resumed BRANCH A	Extractor / operator	After flush stop	App button/decision: Confirm Flow Resumed. If yes, the run advances to final purge.
22B	Branch B - flow not resumed; adjust and re-check BRANCH B	Extractor / operator; supervisor if repeated issue	Until flow resumes	If flow has not resumed, the app enters Flow Adjustment Required and offers Re-check Flow. Repeat adjustments and decision until the flow decision is yes.
Final purge, clarity, and booth shutdown				
23	Start final purge APP	Extractor / operator	After flow resumed is confirmed	App button: Start Final Purge. Optional target duration can be configured in Settings; default policy is informational unless tightened.
24	Stop final purge APP	Extractor / operator	When purge/burp is complete	App button: Stop Final Purge. Duration is computed from start/end timestamps.
Branch C/D - final clarity decision				

STEP	NAME OF PROCESS	PERSON / ROLE RESPONSIBLE	TIME	WHAT HAPPENS / CRITICAL NOTES
25C	Branch C - final clarity confirmed BRANCH C	Extractor / operator	After final purge	App button/decision: Confirm Final Clarity. If yes, the process can move to shutdown checklist.
25D	Branch D - more purge / clarity work required BRANCH D	Extractor / operator; supervisor if repeated issue	Until clear enough to proceed	If not clear enough, the app enters More Purge / Clarity Work Required and offers Resume Final Purge. Repeat final purge and clarity confirmation until clear.
26	Complete shutdown checklist CRITICAL	Extractor / operator	After final clarity is confirmed	App checklist fields: Recovery inlets closed, Filtration pump-down started, Nitrogen off, Dewax inlet closed. These are captured as shutdown timestamps before closing the booth process.
27	Mark run complete APP	Extractor / operator	After shutdown checklist	App button: Mark Run Complete. This records run completion and completes the linked reactor event when it is still active.

Phase 3 - Wet THCA / HTE Pour And Handoff

Initial output capture and immediate downstream handoff

STEP	NAME OF PROCESS	PERSON / ROLE RESPONSIBLE	TIME	WHAT HAPPENS / CRITICAL NOTES
28	Select post-extraction pathway HANDOFF	Extractor or supervisor	After run completion; before downstream starts	App choices: 100 lb pot pour or 200 lb minor run. The current question stops at the initial THCA/HTE pour, but the pathway should be selected so downstream knows what workflow owns the material next.
29	Start post-extraction handoff HANDOFF	Extractor / downstream operator	Immediately before pour/output capture	App button: Start Post-Extraction. This is blocked until the extraction run is complete.
30	Prepare receiving containers and labels INFERRED	Extractor / downstream operator	Before opening/pouring outputs	Prepare labeled containers for wet THCA and wet HTE. The app captures weights but does not currently enforce container labeling, tare, or label scan steps in this exact handoff.
31	Pour / collect wet HTE CRITICAL	Extractor / downstream operator	During output pour	Collect wet HTE into the correct receiving container. Record Wet HTE in grams in the app before confirming initial outputs.
32	Pour / collect wet THCA CRITICAL	Extractor / downstream operator	During output pour	Collect wet THCA into the correct receiving container. Record Wet THCA in grams in the app before confirming initial outputs.
33	Confirm initial outputs APP	Extractor or supervisor	After both wet output weights are entered	App button: Confirm Initial Outputs. This requires both Wet HTE and Wet THCA. It timestamps the handoff and allows the chosen downstream branch to continue.
34	Clean / reset extraction equipment for next run INFERRED	Extractor; maintenance/facilities as assigned	After output pour and shutdown, before next charge	Clean tanks, lines, reactor, filtration path, and receiving area according to SOP. The current app does not make cleaning a first-class completion gate, but production feedback suggests it should be visible in the operator workflow.

Suggested Additions To Make This Easier To Follow

- Add a one-page laminated "Run Card" with the same step numbers as the app buttons.
- Group the tablet workflow into four big phases: Charge, Primary, Flush/Purge, Pour/Handoff.
- Add a visual progress rail with Branch A/B and Branch C/D loops shown directly on screen.
- Make prep and cleaning visible: reactor ready, solvent tank ready, containers labeled, post-run clean/reset.
- Use role initials on each step: EXT, SUP, DOWN, MAINT.

Process Gaps Worth Confirming With The Team

- Exact owner and timing for tank cleaning, solvent loading, and reactor readiness checks.
- Whether -40 F is a hard stop or a supervisor-overridable warning in production.
- Whether Slack evidence remains required or app photo evidence replaces it as the source of truth.
- Whether wet HTE and wet THCA should require container labels, tare weights, photos, or supervisor signoff.
- Whether run complete should mean booth shutdown complete or output pour complete.